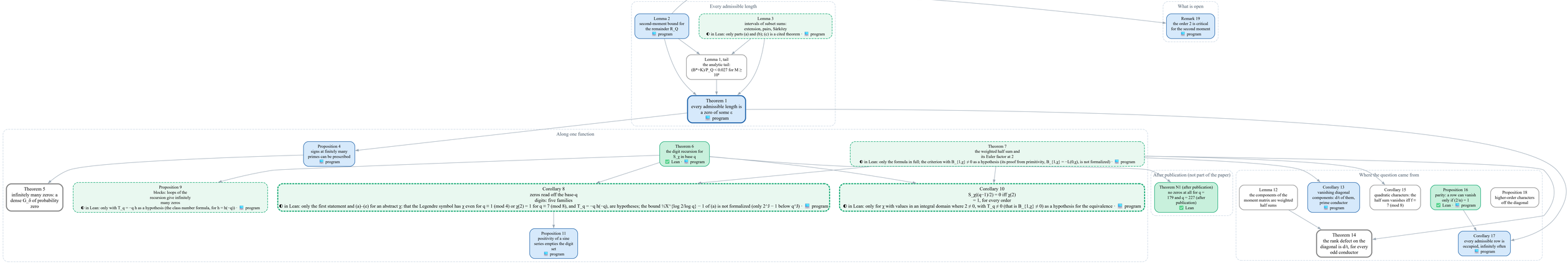
Map of *Zeros of weighted partial sums of completely multiplicative functions: Dirichlet characters, the Euler factor at 2 and Conrey's sine series*

Number theory · arithmetic functions

21 results | **3** formalized in Lean | **5** with a part in Lean | **14** checked by a program | written proof only | (a result can carry several tags)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



What the paper contributes

For a completely multiplicative $\varepsilon : \mathbb{N} \to \{\pm 1\}$, the weighted sum $S_\varepsilon(m) = \Sigma_{n \le m}\, n\, \varepsilon(n)$ can vanish only when $m \equiv 0, 3 \pmod 4$.

- **Every admissible length is a zero** (Theorem 1): for every such $m \ge 3$ some ε has $S_\varepsilon(m) = 0$ — a second moment, 104 integer certificates up to 10^8, and Sárközy's progressions in subset sums beyond.
- **One fixed function rarely vanishes, but generically does** (Theorem 5): the ε with infinitely many zeros form a dense G_δ of probability zero, since every finite prefix of signs extends to a zero (Proposition 4).
- **Quadratic characters are explicit points of that set**: a digit recursion in base q (Theorem 6) propagates their zeros for $q \equiv 1 \pmod 4$, for $q \equiv 7 \pmod 8$ through the Euler factor at 2 of the weighted half sum (Theorem 7, classical in form), for class number one and along loops of the recursion (Corollary 8, Proposition 9); for odd characters of any order at $(q^j - 1)/2$ exactly when $\chi(2) = 1$ (Corollary 10).
- **The digits are Conrey's null cells**: they are the cells on which Conrey's sine series vanishes identically; positivity of a related series would exclude them for $q \equiv 3 \pmod 8$ (Proposition 11).
- **The origin**: on the diagonal of the moment matrix of the ring of character values, the vanishing components are the odd characters trivial on $\langle 2 \rangle$, and the rank defect is d/t for every odd conductor (Theorem 14, known in substance: Jager–Lenstra, Girstmair).
- **Added after publication**: for $q = 179$ and 227, $\chi_{q,1}$ has no zero at all (N1), by a finite certificate checked in Lean.

🚀 **Start here: Theorem 1**: every admissible length is a zero of some ε.

Main results:

- **Theorem 1**: every admissible length is a zero of some ε
- **Theorem 5**: infinitely many zeros: a dense G_δ of probability zero
- **Corollary 8**: zeros read off the base-q digits: five families
- **Corollary 10**: $S_\chi((q-1)/2) = 0$ iff $\chi(2) = 1$, for every order
- **Theorem 14**: the rank defect on the diagonal is d/t, for every odd conductor

🧭 Fields: where the paper lives and what it connects

Tools taken from other branches

- **Additive combinatorics** → arithmetic progressions in subset sums (Sárközy, with Lev's constants in the form of Bringmann–Wellnitz); complete sequences and pairs — Theorem 1, Lemma 3
- **Probability** → a second-moment bound; the anticoncentration theorem of Sárközy and Szemerédi; Borel–Cantelli — Lemma 2, Theorem 5
- **Topology (Baire category)** → dense G_δ sets in the product space of sign patterns — Theorem 5
- **Algebraic number theory** → generalized Bernoulli numbers $B_{1,\chi}$, the class number formula of Jacobi and Dirichlet, Euler factors — Theorem 7, Corollary 8, Corollary 15
- **Representation theory of finite abelian groups** → the idempotents e_χ of $\mathbb{C}[G]$: the rank of a cyclic module is the number of non-vanishing Fourier components — Lemma 12, Theorem 14

- **Formal verification** → Lean 4 + Mathlib: the digit recursion and its corollaries, the weighted half sum, the subset-sum lemmas, the parity of the rows, and the finite certificate of N1 — Theorem 6, Corollary 8, Theorem 7, Lemma 3, Proposition 16, Theorem N1 (after publication)

What the results serve

- **Probabilistic number theory** ← random multiplicative functions: with the weight n the sum jumps over 0, against the unweighted case (Aymone) — Theorem 1, Theorem 5
- **Analytic number theory (character sums)** ← Conrey's sine series of the Legendre symbol, from his approach to the Riemann hypothesis: its null cells are the digits of the recursion — Corollary 8, Proposition 11
- **Partial sums of Legendre symbols** ← the digit recursion of Borwein–Choi–Coons and Klurman–Mangerel–Pohoata–Teräväinen, with the weight added — Theorem 6, Corollary 8
- **Rings of character values (the companion paper)** ← the components and the rank of the moment matrix on its diagonal, and which rows off it can be occupied — Lemma 12, Theorem 14, Corollary 17